Kenya
Competition Authority of Kenya

Businesses continue to leverage information and communication technology tools globally, especially through digitization, in order to be more efficient and reliable in their processes.[29] This brief demonstrates how integration of customer relationship management ('CRM') and enterprise resource planning ('ERP') systems have improved operational efficiency in competition law enforcement in Kenya. These systems have become key in storage, aggregation and referencing of agency-wide records and information, and as a result reduced filing costs and turnaround time by up to 50 per cent. The installation and management of the systems has enabled uninterrupted service delivery in the face of adversity and shocks associated with the global COVID-19 outbreaks. This brief provides a testimony of how competition agencies can take advantage of investment in and installation of technology-enhanced effectiveness and improve the operational efficiency of their regulatory efforts. A key insight is that upskilling human resources is critical for the implementation of computational tools. Furthermore, the adoption of such tools is also impacted by the recent development of data protection and privacy laws in the digital economy landscape. The data and information aggregated through digitization allow for evidentiary, coherent and predictable competition law enforcement.

I. The automation of complaints/ case processing

In recent years, businesses globally have leveraged information and communication technology tools in order to be more efficient and reliable in their business processes. The COVID-19 pandemic heightened the need to enhance the use of digital tools, including in the context of the detection, prevention and deterrence of anticompetitive conduct by competition agencies.

The Government of Kenya clearly notes the potential of digitization to boost the growth of African economies. The need for a regulatory framework that facilitates digitization stems from the fact that digitization boosts investments, enables product innovation and the development of data-based services.[30] In this regard, the

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[30] The Republic of Kenya, *Digital Economy Blueprint: Powering Kenya's Transformation*,

Competition Authority of Kenya ('the Authority') cognizant of the need to enhance efficiency of its operation, for both internal and external stakeholders, operationalized the Case Management System ('CMS') in 2020, after almost four years from conceptualization. The CMS has automated the processing of matters related to merger filings, restrictive trade practices and abuse of buyer power cases, as well as consumer protection complaints.

The agency's digitization process commenced in 2018 when the Authority converted all its paper files into digital records and archived them in relevant streams in the CMS and integrated them with the Document Management System ('DMS') for easy referencing. The digitalization process has also enabled the case handlers to access the system remotely, and has rendered compliance with the data protection laws easily attainable.[31]

The introduction of electronic DMS records was a significant milestone since any undertaking's submission received in hardcopy is now scanned and uploaded by the Authority's registry personnel to be assigned to a case handler for processing. The CMS and DMS systems completed the cycle by facilitating undertakings/stakeholders to upload digital submissions remotely, from any geographical location both nationally and globally, through the Authority's Online filing ('E-Filing') Portal.[32] Filings made through this online platform are automatically assigned to CMS files and the registry officials are notified. Further, any attachments, including evidentiary information if it pertains to a complaint or merger application form from external parties, are automatically captured in the system.

This process also ensured the full digitization of the entire Authority's operations with the support of functional units (departments) through the enterprise resource planning ('ERP') system. This system caters for, among others, human resource productivity monitoring, procurement, electronic submission and review of job applications, application of various types of leave, and non-technical processes that are critical for the smooth functioning of the Authority.

https://www.ict.go.ke/wp-content/uploads/2019/05/Kenya-Digital-Economy-2019.pdf (last accessed June 11, 2023).

[31] The National Council of Law Report, *The Data Protection Act No. 24 of 2019*, https://www.odpc.go.ke/download/kenya-gazette-data-protection-act-2019/?wpdmdl=3235&refresh=64834673538de1686324851 (last accessed June 11, 2023).

[32] Competition Authority of Kenya, *E-filing Portal* https://competition.cak.go.ke:444/ (last accessed June 11, 2023).

The CMS and the DMS systems are comprised of (dashboard and workflows) that have the following main features:

- The DMS contains the record of all saved files and their exact location with an instantaneous search engine;
- There is a file management module integrated with an administration and information upload interface used by the Registry Team;
- CMS interfaces are used for instructing case handlers (legal and economic analysts) to determine the progress of each file assigned to them;
- The local control interface enables the Authority's senior Management to be notified of all files assigned to their domain, their status, and case officers interacting with them;
- The general control interface enables staff to know all the files assigned to each department of the Authority, the progress, and case officers who interact with them;
- Control users have an overview that includes the following information and functions:
    - Real-time reports which include the cases handed down in specific number of days and communicated to relevant staff;
    - Flagging case files whose processing dates are maturing and due to expiration or require urgent action;
    - An overview of files initiated at the Registry, officers handling them at the time and their review status;
    - The number of cases in process, classified into those with a final decision or resolved, and those that are still being processed;
    - The timeline of each system file, with details of the stages it has gone through during the processing and the respective dates and duration of each stage.

The CMS and the ERP systems have enabled faster operations and seamless contact with the stakeholders and customers. Filing for a Merger, for instance, before the implementation of the current systems, involved the applicant printing documents, binding and hand-delivering them via ordinary post to the Authority, within the official working hours. This was not only costly, but also time-consuming. Following the automation of the Authority's services, stakeholder feedback indicates that filing costs and turnaround time have been reduced by up to 50%. Applications can be made round the clock since September 2019. Today, all that a party or stakeholder needs is access to internet from any part of the world to file a merger.

In promoting transparency in public procurement, the Authority has embraced e-procurement, whereby bidders submit their bids online and can track their progress. The Authority's e-procurement portal has further enriched customer experience by providing an interactive system where surveys are conducted in a rather cost-effective manner that would otherwise have been costly and involved inconveniences.

In a nutshell, the Authority has saved 90 per cent of operational costs due to digitization, including printing and human capital. The turnaround time was reduced from five to two days, increasing efficiency and speed. Internally, processes such as financial and human resource management have also been automated, resulting in efficiencies such as shorter turnaround time in attending to matters concerning both external and internal stakeholders. In summary, the deployment of the CMS and DMS systems have enabled the Authority execute its mandate in a more cost-efficient manner and facilitated a faster pace for the resolution of cases, thereby enhancing customer satisfaction and raising the profile of the Authority as a well-functioning Government agency.

II. The use of digital tools to detect, investigate and deter anticompetitive practices

The proactive digital tools used by the Authority consist of installations that are relied on in cartel investigations, in accordance with the provisions of the Competition Act No. 12 of 2010.[33] Furthermore, the enforcement of public procurement laws is also digitized in order to ensure that the goods/services procured are adequately priced and of good quality.

In order to enhance the quality and potency of its cartel investigations, the Authority has competitively contracted external digital consultants with specific hardware and software skills and knowledge. Digital tools are relied on in the investigative process to collect, process, and analyze digital evidence, therefore improve data gathering during dawn raids and the subsequent data analysis. The use of such modern tools facilitates the efficient analysis of data running into tens of terabytes. The software further facilitates the generation of reports that enable better interpretation of the evidence resulting from the data collected.

[33] The Republic of Kenya, *The Competition Act of Kenya No. 12 of 2010*, https://www.cak.go.ke/sites/default/files/Competition_Act_No._2012_of_2010.pdf (last accessed June 11, 2023).

The hardware used by forensic experts, in liaison with the Authority's staff, include TX1 Forensic Imagers which greatly reduce the time taken to develop forensic images of retrieved data. Further, the Detego Forensic Imager and Analyzer permits access to and reading of the data in storage devices without compromising the integrity of the data. The analyzer has the capability to read from various storage media (hard drives, USB and servers) that contain the digital evidence. It includes writing blockers to guarantee the protection and integrity of the files subject to analysis, thereby preserving the chain of custody of evidence. The Authority also employs an FTK Analysis Toolkit with the capacity of forensic imaging of data in storage devices, either as one file or in segments that may be reconstructed later for analysis.

Additional equipment deployed by the Authority's external consultants in recent dawn raids (cartel investigations) includes the Fred's Mobile Phone Data Extractor. The tool, which the Authority made use of for the first time in December 2021, has proven useful for assisting in the search for keywords that point to concerted practices among competitors, thereby helping build a strong case against the undertakings under investigation. The extractor also facilitated the acquisition of relevant information from the mobile phones of key suspects. The Authority is in the process of procuring these tools in order to build an in-house forensics laboratory and expertise, in order to cut cost and at the same time build in-house competency of the internal users. These tools are expected to improve the Authority's ability to obtain and analyze data pertaining to the investigation of anti-competitive practices. Further, case-handlers have indeed received training from various forensic experts and are therefore able to competently deploy the tools to achieve the desired objectives of data retrieval and analysis.

In collaboration with the public procurement regulator—Public Procurement Regulatory Authority ('PPRA')—the Authority is in the process of developing a toolkit for the detection of bid rigging through the analysis of key information, and this entails uploading tender documentation.[34] Using algorithms, the investigation tool will detect red flags in key areas of potential concern such as the number of bidders, pricing patterns, document origin and staggered bids. The existence of a red flag, or a combination of several indicators, could signal bid-rigging, thereby warranting a closer review of the applications, including physical assessment. Bids

[34] The Republic of Kenya, *The Data Protection Act No. 24 of 2019*, https://www.odpc.go.ke/download/kenya-gazette-data-protection-act-2019/?wpdmdl=3235&refresh=64834673538de1686324851 (last accessed June 11, 2023).

that score highly in this scale are immediately flagged for more advanced investigations.

The collaborative partnership with the public procurement regulator is borne of the fact that the public procurement of goods and services accounts for approximately 40 per cent of the Government of Kenya's budget annually. Therefore, efficiency in this sector is key to converting fiscal plans into results. Once successfully completed, this project will enable the Authority to detect and resolve allegations of bid rigging with relative ease, thus increasing the scope of surveillance of public procurement. The net effect of bid rigging is that purchasers--national and regional Governments--end up paying much more than the competitive price for services whose quality is often not guaranteed. Collusive tendering usually also stifles innovation.

III. Use of Digital Tools to Increase Agency Effectiveness

In relation to the analysis of market inquiry data, like the concluded Digital Credit Market Inquiry, the Authority used an open source data analysis tool--R Analytics. This software was cost effective and provided an opportunity for analysts to master various commands and stay up to date with updates and new modules with expanded versatility. However, the downside of this software is that it has limits on the volumes of data that can be processed. This is unlike software such as the statistical program for social sciences ('SPSS') which can process considerably larger volumes.

The Authority has also incorporated open source Knowledge Management tools to manage both the resource center and the digital content. These are KOHA and D-Space. KOHA, which is utilized by over 3,000 libraries across the world, has given the Authority access to a wide array of relevant and up-to-date resources and information from the library. On the other hand, D-Space preserves and supports easy and open access to digital content including text, images, moving images, .mpegs and data sets available within the Authority. All Authority staff have access to the data repository that supports the development of various reports and advisories.

IV. Conclusion

The relevance of technology for competition enforcement has become evident for the Authority, therefore the investment in the development and implementation of a case management system, acquisition of digital forensic tools and usage of off-the-shelf software are paying off. It is not only about being able to understand how disruptive technologies work but also to fulfill the legal mandate of the Authority effectively and efficiently by drawing on resources and tools that support priority activities and enhance the timely resolution of cases.

The use of computational tools like algorithms and artificial intelligence are still in their nascent stages of use by competition agencies in developing countries. However, the Authority takes a proactive approach in upskilling its case handlers regarding the latest technology, knowledge and practical skills to ensure the officers are able to keep pace with the dynamism and rapidly changing the face of competition law enforcement. Mindful of this, the Authority aspires to learn from the expertise of other agencies through the Stanford Computational Antitrust Project.